

C. STAAD INPUT DATA :

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*****
*          STAAD.Pro          *
*          Version 2007      Build 01          *
*          Proprietary Program of              *
*          Research Engineers, Intl.            *
*          Date=    OCT 13, 2008              *
*          Time=    16: 7:58                  *
*          USER ID:  ENGINEERING              *
*****
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1. STAAD SPACE
INPUT FILE: SKYLIGHT-FRAME_STEEL-FINAL2.STD
2. START JOB INFORMATION
3. ENGINEER DATE 09-OCT-08
4. JOB NAME SKYLIGHT FRAME
5. JOB CLIENT DMP. (RASHIDIYA)
6. JOB NO 001
7. JOB REV A2
6. JOB REF DM001-D-MDF-CAR-SD-DCC-1230101REV2
9. JOB COMMENT PRELIMINARY ANALYSIS
10. ENGINEER NAME AA
11. END JOB INFORMATION
12. INPUT WIDTH 79
13. UNIT MMS NEWTON
14. JOINT COORDINATES
15. 1 0 -1314.91 25090.1; 2 0 -101.147 1930; 3 0 -202.294 3860; 4 0 -303.441 5790
16. 5 0 -404.588 7720.01; 6 0 -505.735 9650.01; 7 0 -606.883 11580
17. 8 0 -708.032 13510.1; 9 0 -809.177 15440; 10 0 -910.326 17370.1
18. 11 0 -1011.47 19300; 12 0 -1112.62 21230.1; 13 0 -1213.77 23160; 14 2115 0 0
19. 15 2115 -1314.91 25090.1; 16 2115 -101.147 1930; 17 2115 -202.294 3860
20. 18 2115 -303.441 5790; 19 2115 -404.588 7720.01; 20 2115 -505.735 9650.01
21. 21 2115 -606.883 11580; 22 2115 -708.032 13510.1; 23 2115 -809.177 15440
22. 24 2115 -910.326 17370.1; 25 2115 -1011.47 19300; 26 2115 -1112.62 21230.1
23. 27 2115 -1213.77 23160; 28 2115 -1359.77 25946; 29 4110 0 0
24. 30 4110 -1314.91 25090.1; 31 4110 -101.147 1930; 32 4110 -202.294 3860
25. 33 4110 -303.441 5790; 34 4110 -404.588 7720.01; 35 4110 -505.735 9650.01
26. 36 4110 -606.883 11580; 37 4110 -708.032 13510.1; 38 4110 -809.177 15440
27. 39 4110 -910.326 17370.1; 40 4110 -1011.47 19300; 41 4110 -1112.62 21230.1
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42. 84 9083 -505.735 9650.01; 85 9602 -606.883 11580; 86 10120 -708.032 13510.1
43. 87 10638 -809.177 15440; 88 11157 -910.326 17370.1; 89 11675 -1011.47 19300
44. 90 11800 -1112.62 21230.1; 91 12193 -1112.62 21230.1; 92 12710 -1213.77 23160
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46. 96 2115 -1001.2 3818.13; 97 2115 -1102.34 5748.14; 98 2115 -1203.49 7678.14
47. 99 2115 -1304.64 9608.14; 100 2115 -1405.79 11538.1; 101 2115 -1506.94 13468.2
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49. 104 2115 -1810.37 19258.1; 105 2115 -1911.52 21188.2
50. 106 2115 -2012.67 23118.1; 107 4110 -2113.82 25048.2
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 71. 150 11800 -2012.67 23118.1; 151 11800 -1911.52 21188.2

72. MEMBER INCIDENCES

73. 1 1 15; 2 2 16; 3 3 17; 4 4 18; 5 5 19; 6 6 20; 7 7 21; 8 8 22; 9 9 23
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132. 388 40 118; 389 117 41; 390 55 131; 391 130 56; 392 66 141; 393 140 67
133. 394 73 147; 395 146 74; 396 26 106; 397 105 27; 398 41 119; 399 118 42
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135. 406 90 150; 407 151 78; 408 27 94; 409 106 15; 410 42 107; 411 119 30
136. 412 57 120; 413 132 45; 414 68 133; 415 142 59; 416 75 143; 417 148 70
137. 418 78 149; 419 150 77
138. START GROUP DEFINITION
139. MEMBER
140. _73DIA_PIPE 133 134 136 138 140 142 144 146 147 149 151 153 155 157 159 160 -
141. 162 164 166 168 170 172 174 176 178 180 182 184 186 188 190 254 TO 257 262 -
142. 263 TO 266 272 TO 279 283 TO 285 289 290 293 294
143. _141.3DIA_PIPE 1 2 4 6 8 10 12 28 29 31 33 35 37 39 55 56 58 60 62 64 66 82 -
144. 84 86 88 90 102 104 106 114 118 120 122 124 126 129 130 132 191 TO 215 217 -
145. 219 221 223 225 227 228 230 232 234 236 238 240
146. _48DIA_PIPE_STRUT 135 137 139 141 143 145 148 150 152 154 156 158 161 163 165 -
147. 167 169 171 173 175 177 179 181 183 185 187 189
148. _36DIA_T_ROD 3 5 7 9 11 13 TO 27 30 32 34 36 38 40 TO 54 57 59 61 63 65 67 -
149. 68 TO 81 83 85 87 89 91 TO 101 103 105 107 TO 113 115 TO 117 119 121 123 125 -
150. 127 128 131
151. _32DIA_T_ROD 216 218 220 222 224 226 229 231 233 235 237 239 241 TO 253 258 -
152. 259 TO 261 267 TO 271 280 TO 282 286 TO 288 291 292 295 TO 419
153. END GROUP DEFINITION
154. DEFINE MATERIAL START
155. ISOTROPIC STEEL
156. E 190000
157. POISSON 0.3
158. DENSITY 7.68195E-005
159. ALPHA 1.2E-005
160. DAMP 0.03
161. END DEFINE MATERIAL
162. MEMBER PROPERTY BRITISH
163. 133 134 136 138 140 142 144 146 147 149 151 153 155 157 159 160 162 164 166 -
164. 168 170 172 174 176 178 180 182 184 186 188 190 254 TO 257 262 TO 266 272 -
165. 273 TO 279 283 TO 285 289 290 293 294 TABLE ST PIPE OD 73 ID 62.68
166. 1 2 4 6 8 10 12 28 29 31 33 35 37 39 55 56 58 60 62 64 66 82 84 86 88 90 102 -
167. 104 106 114 118 120 122 124 126 129 130 132 191 TO 215 217 219 221 223 225 -
168. 227 228 230 232 234 236 238 240 TABLE ST PIPE OD 141.3 ID 125.3
169. 135 137 139 141 143 145 148 150 152 154 156 158 161 163 165 167 169 171 173 -
170. 175 177 179 181 183 185 187 189 TABLE ST PIPE OD 148.5 ID 125.3
171. 3 5 7 9 11 13 TO 27 30 32 34 36 38 40 TO 54 57 59 61 63 65 67 TO 81 83 85 -
172. 87 89 91 TO 101 103 105 107 TO 113 115 TO 117 119 121 123 125 127 128 -
173. 131 TABLE ST PIPE OD 36 ID 0
174. 216 218 220 222 224 226 229 231 233 235 237 239 241 TO 253 258 TO 261 267 -
175. 268 TO 271 280 TO 282 286 TO 288 291 292 295 TO 418 -
176. 419 TABLE ST PIPE OD 32 ID 0

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177. CONSTANTS
178. MATERIAL STEEL ALL
179. SUPPORTS
180. 1 TO 14 28 29 43 44 58 69 76 79 TO 89 91 TO 93 FIXED
181. MEMBER TENSION
182. 3 5 7 9 11 13 14 27 41 54 68 81 101 113 117 119 121 123 125 127 131 216 218 -
183. 220 222 224 226 229 231 233 235 237 239 241 TO 249 304 305 308 309 380 391
184. MEMBER TRUSS
185. 250 TO 253 258 TO 261 267 TO 271 280 TO 282 286 TO 288 291 292 295 TO 303 -
186. 306 307 310 TO 379 381 TO 390 392 TO 419
187. LOAD 1 LOADTYPE MASS TITLE SELF WEIGHT OF STRUCTURE
188. SELFWEIGHT Y -1
189. LOAD 2 LOADTYPE DEAD TITLE DEAD WEIGHT OF GLASS
190. JOINT LOAD
191. 16 TO 27 FY -3356.25
192. 31 TO 42 50 TO 57 FY -3278.99
193. 46 FY -2371.26
194. 47 FY -2794.01
195. 48 FY -3216.76
196. 49 FY -3354.1
197. 60 FY -2000.01
198. 61 FY -2422.76
199. 62 FY -2847.66
200. 63 FY -3272.55
201. 64 FY -3321.91
202. 71 FY -2055.81
203. 65 TO 68 FY -3158.82
204. 72 FY -2478.56
205. 73 FY -2901.31
206. 74 FY -3075.13
207. 90 FY -1804.73
208. 75 FY -3038.65
209. 78 FY -2229.63
210. 15 FY -2693.15
211. 30 45 FY -2630.92
212. 59 FY -2534.35
213. 70 FY -2437.79
214. 77 FY -2062.25
215. LOAD 3 LOADTYPE WIND TITLE LOAD @ SPIDER DUE TO POS-WIND LOAD
216. JOINT LOAD
217. 16 TO 27 FY -5943.2
218. 31 TO 42 50 TO 57 FY -5806.4
219. 46 FY -4199
220. 47 FY -4947.6
221. 48 FY -5696.2
222. 49 FY -5939.4
223. 60 FY -3541.6
224. 61 FY -4290.2
225. 62 FY -5042.6
226. 63 FY -5795
227. 64 FY -5882.4
228. 71 FY -3277.91
229. 65 TO 68 FY -5593.6
230. 72 FY -3457.79
231. 73 FY -3795.99
232. 74 FY -4672.51
233. 90 FY -2181.26
234. 75 FY -5380.9
235. 78 FY -3258.64
236. 15 FY -4769
237. 30 45 FY -4658.8
238. 59 FY -4487.8
239. 70 FY -4316.8

240. 77 FY -2833.08
241. LOAD 4 LOADTYPE WIND TITLE LOAD @ SPIDER DUE TO NEG-WIND LOAD
242. JOINT LOAD
243. 16 TO 27 FY 11417.2
244. 31 TO 42 50 TO 57 FY 11154.4
245. 46 FY 8066.5
246. 47 FY 9504.6
247. 48 FY 10942.7
248. 49 FY 11409.9
249. 60 FY 6803.6
250. 61 FY 8241.7
251. 62 FY 9687.1
252. 63 FY 11132.5
253. 64 FY 11300.4
254. 71 FY 5424.49
255. 65 TO 68 FY 10745.6
256. 72 FY 4401.01
257. 73 FY 4062.81
258. 74 FY 7115.69
259. 90 FY 1748.14
260. 75 FY 10336.8
261. 78 FY 4600.16
262. 15 FY 9161.5
263. 30 45 FY 8949.8
264. 59 FY 8621.3
265. 70 FY 8292.8
266. 77 FY 3471.72
267. LOAD COMB 5 (SLF + DL + POS_WL)_NOMINAL
268. 1 1.0 2 1.0 3 1.0
269. LOAD COMB 6 (SLF + DL + NEG_WL)_NOMINAL
270. 1 1.0 2 1.0 4 1.0
271. LOAD COMB 7 (SLF + DL + POS_WL)_FACTORED
272. 1 1.2 2 1.2 3 1.2
273. LOAD COMB 8 (SLF + DL + NEG_WL)_FACTORED
274. 1 1.2 2 1.2 4 1.2
275. PERFORM ANALYSIS
276. PRINT MAX. JOINT DISPLACEMENT
JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
56	5	-0.0188	-0.9813	0.0481	-0.0003	-0.0001	0.0000

277. PRINT SECTION DISPL

MEMBER SECTION DISPLACEMENTS

UNIT = INCHES FOR FPS AND CM FOR METRICS/SI SYSTEM

MEMB	LOAD	GLOBAL X,Y,Z DISPL FROM START TO END JOINTS AT 1/12TH PTS					
66	5	-0.0062	-0.8114	0.0301	-0.0072	-0.8344	0.0312
		-0.0083	-0.8566	0.0323	-0.0093	-0.8776	0.0335
		-0.0104	-0.8972	0.0347	-0.0114	-0.9152	0.0361
		-0.0125	-0.9314	0.0375	-0.0135	-0.9457	0.0390
		-0.0146	-0.9577	0.0406	-0.0156	-0.9675	0.0423
		-0.0167	-0.9747	0.0441	-0.0177	-0.9794	0.0460
		-0.0188	-0.9813	0.0481			

MAX LOCAL DISP = 0.03519 AT 116.38 LOAD 5 L/DISP= 5668